

CSCI 5244: Homework Set 2

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1. Recall Pauli X,Y,Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(a) Show that for any $j \in \{1, 2, 3\}$

$$e^{-i\theta\sigma_j} = I \cos(\theta) - i\sigma_j \sin(\theta)$$

It is known that e^x can expand into the Taylor series

$$e^x = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

For Pauli matrices X, Y, Z, when powers are even they all become the identity matrix I :

$$X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$Y^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$Z^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

If we take the cubic power for these matrices they will all turn back into themselves:

$$\begin{aligned} X^3 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ Y^3 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ Z^3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

So putting in $x = i\theta\sigma_j$ we have

$$\begin{aligned} e^{-i\theta\sigma_j} &= 1 - i\theta\sigma_j + \frac{(i\theta\sigma_j)^2}{2!} - \frac{(i\theta\sigma_j)^3}{3!} + \frac{(i\theta\sigma_j)^4}{4!} \\ &= 1 - i\theta\sigma_j - I \frac{\theta^2}{2!} + i\sigma_j \frac{\theta^3}{3!} + I \frac{\theta^4}{4!} \dots \end{aligned}$$

using how the $X^2, Y^2, Z^2 = I$ and $i^2 = -1, i^3 = -i, i^4 = 1$, etc.

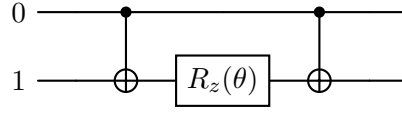
Recall that $i^2 = -1, i^3 = -i, i^4 = 1$. The Taylor series expansion of sin and cos are

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \text{ and } \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

So

$$\begin{aligned} e^{-i\theta\sigma_j} &= \left(1 - I\frac{\theta^2}{2!} + I\frac{\theta^4}{4!} - \dots\right) - i\left(\theta - \sigma_j\frac{\theta^3}{3!} + \sigma_j\frac{\theta^5}{5!} - \dots\right) \\ &= I \cos \theta - i\sigma_j \sin \theta \end{aligned}$$

- (b) Consider a system of two qubits. Denote Z_0 and Z_1 the Pauli Z matrices on qubits 0 and 1, respectively. Show that the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$ can be implemented by the following circuit with $R_z(\theta) = e^{-i\theta\sigma_3}$



We want to show that this circuit implements the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$.

Since $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$ for any basis state $|x_0 x_1\rangle$

$$Z_0 Z_1 |x_0 x_1\rangle = (-1)^{x_0 \otimes x_1} |x_0 x_1\rangle$$

Using $R_z(\theta) = I \cos(\theta) - i\sigma_3 \sin(\theta)$ from part (a), we get $R_z(\theta)|0\rangle = e^{-i\theta}|0\rangle$ and $R_z(\theta)|1\rangle = e^{i\theta}|1\rangle$. Below is the trace of the basis state through each gate:

Input	After CNOT	After $R_z(\theta)$	After CNOT
$ 00\rangle$	$ 00\rangle$	$e^{-i\theta} 00\rangle$	$e^{-i\theta} 00\rangle$
$ 01\rangle$	$ 01\rangle$	$e^{i\theta} 01\rangle$	$e^{i\theta} 00\rangle$
$ 10\rangle$	$ 11\rangle$	$e^{i\theta} 11\rangle$	$e^{i\theta} 10\rangle$
$ 11\rangle$	$ 10\rangle$	$e^{-i\theta} 10\rangle$	$e^{-i\theta} 11\rangle$

In every case the circuit output matches $e^{-i\theta Z_0 Z_1}$ exactly, thus this circuit implements the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$.

2. Amplitude Amplification Given a unitary operator U on a system of n qubits:

$$U|o^{\otimes n}\rangle = \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle$$

where $\langle\Psi_0|\Psi_0\rangle = \langle\Psi_1|\Psi_1\rangle = 1$ and $\langle\Psi_0|\Psi_1\rangle = 0$. Define $S_1 = I - 2|\Psi_1\rangle\langle\Psi_1|$, and $S_i = I - 2|0^{\otimes n}\rangle\langle 0^{\otimes n}|$, and $Q = -US_iU^{-1}S_1$.

- (a) Show that

$$\begin{aligned} Q|\Psi_1\rangle &= \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle \\ Q|\Psi_0\rangle &= \sin 2\theta |\Psi_1\rangle + \cos 2\theta |\Psi_0\rangle \end{aligned}$$

Implies the two-dimensional space spanned by $|\Psi_0\rangle$ and $|\Psi_1\rangle$ is invariant under operator Q .

(b) Show that

$$Q^k U |O^{\otimes n}\rangle = \cos(2k+1)\theta |\Psi_0\rangle + \sin(2k+1)\theta |\Psi_1\rangle, k \in \mathbb{Z}_+$$